Google

SEARCH ANALYSIS PROJECT

Analyze and visualize Google search trends. It presents keyword performance and search insights in a clean and easy-to-understand format, helping transform raw data into meaningful insights for better decision-making.

www.google.com

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OVERVIEW OF GOOGLE

As of recent estimates (2025–2026), Google processes over 16.4 billion searches per day

Google handles more than 190,000 searches every second, which equals:

- ~11.4 million searches per minute
- ~16.4 billion searches per day
- ~5.9 trillion searches per year

Google Trends data represents a very reliable proxy for public interest, making it ideal for:

- Market research
- Political forecasting
- News and media analysis
- Product demand analysis

CREATING QUESTIONS TO SOLVE PROBLEM

If working in a data-driven job role, and you have to answer all these questions.

1) First, write a code where, by changing just the keyword, we can search for multiple things.

2) Top 15 countries where the keywords are searched the most, and also create visual representation of it.

3) A world map needs to be plotted showing the countries that search the keyword the most.

4) We need to extract the time-wise interest of the keyword — how it trended in different years.

5) Compare related keywords and plot the graph.

Analyze search trends for different keywords by allowing easy keyword changes and extracting global search data. Visualize insights using top-15 country charts, a world map, time-wise trends, and comparison graphs of related keywords.

ABOUT OF LIBRARYS TO USE IN PROJECT

Pandas – Data analysis, data manipulation
Matplotlib – Basic data visualization
Seaborn – Statistical visualization
Plotly – Interactive visualizations 📊

Pytrends – Pytrends is an unofficial Python library that allows easy access to Google Trends data through Google's internal API. Since Google doesn't provide an official public API for Trends data, Pytrends acts as a smart wrapper around Google's web interface, enabling data analysts and developers to extract trend data for various keywords, regions, and timeframes.

It is especially useful for:
•Market research
•Trend forecasting
•Competitive analysis
•Keyword research for SEO

IMPORTING LIBRARIES IN PYTHON NOTEBOOK

```
[ ]
1 !pip install pytrends matplotlib pandas seaborn plotly
```

```
Collecting pytrends
  Downloading pytrends-4.9.2-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (5.24.1)
Requirement already satisfied: requests>=2.0 in /usr/local/lib/python3.12/dist-packages (from pytrends) (2.32.4)
Requirement already satisfied: lxml in /usr/local/lib/python3.12/dist-packages (from pytrends) (6.0.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: numpy>=1.23 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly) (9.1.4)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0->pytrends) (3.4.5)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0->pytrends) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0->pytrends) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0->pytrends) (2026.2.25)
Downloading pytrends-4.9.2-py3-none-any.whl (15 kB)
Installing collected packages: pytrends
Successfully installed pytrends-4.9.2
```

```
[ ]
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import seaborn as sns
4 import plotly.express as px
5 from pytrends.request import TrendReq
6 import warnings
7 warnings.filterwarnings("ignore")
```

WORK ON LIBRARYS

Setup pytrends library and keywords define

```
pytrends = TrendReq(hl = 'en-US' , tz = 360)
keyword = 'data science'
```

Data Reuqest

```
pytrends.build_payload([keyword] , cat = 0 , timeframe = 'today 12-m',geo = '',gprop='')
```

Show code

Description of the Code:

Setup pytrends and define keyword :-

This code initializes the Pytrends connection using TrendReq and sets the keyword "data science" to fetch search trend data from Google Trends. 🔎

Data Request (build payload) :-

The build_payload() function prepares the request to Google Trends for the selected keyword with no category filter, global location, and last 12 months timeframe to collect search interest data. 📊🌍

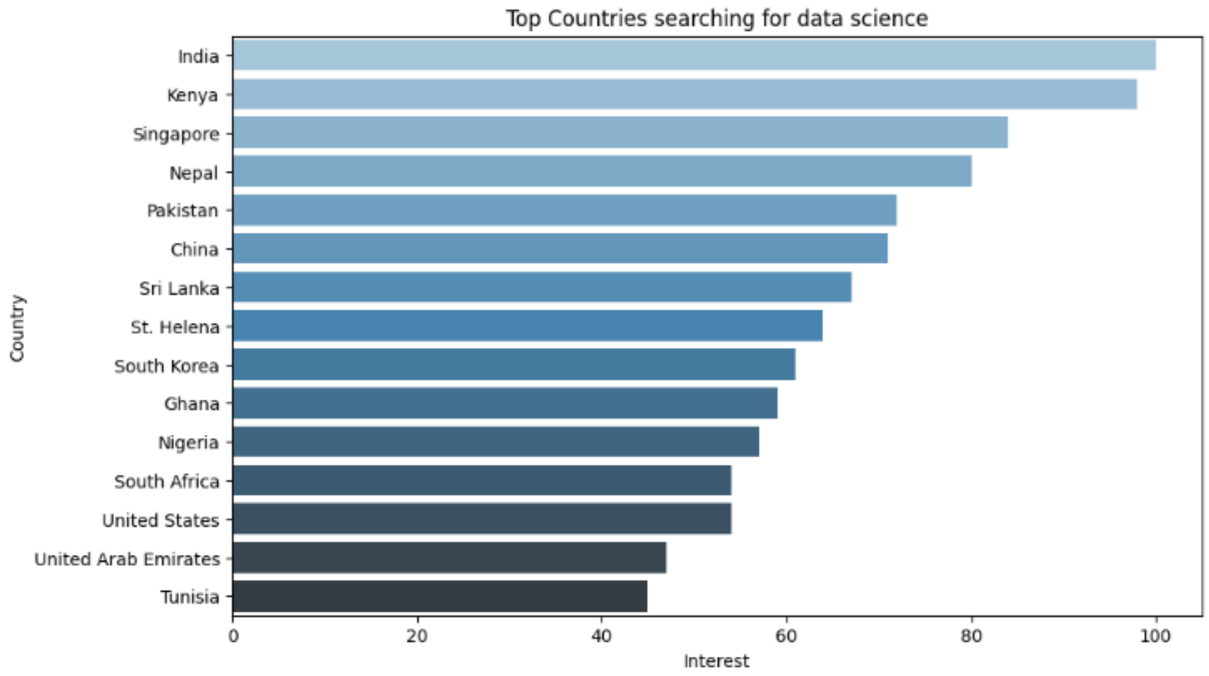
WORK ON LIBRARYS

Country Wise Interest

```
region_data = pytrends.interest_by_region()
region_data = region_data.sort_values(by = keyword , ascending = False).head(15)

```

```
plt.figure(figsize = (10,6))
sns.barplot(x = region_data[keyword] , y = region_data.index ,data = region_data  , palette= "Blues_d")
plt.title(f'Top Countries searching for {keyword}')
plt.xlabel('Interest')
plt.ylabel('Country')
plt.show()
```



Data Extraction: Collect country-wise search interest for the keyword and select the top 15 countries with the highest searches.

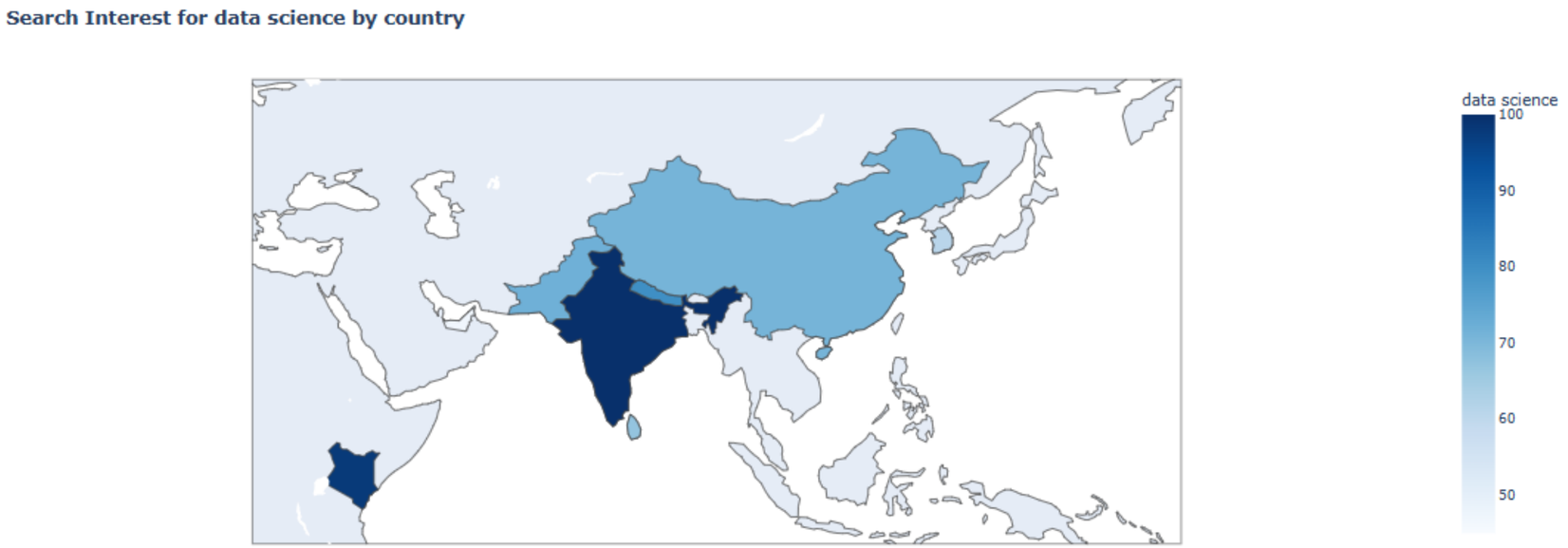
Data Visualization: Display the results using a horizontal bar chart to show which countries search the keyword the most. 📊🌍

WORK ON LIBRARYS

World Map

```
region_data = region_data.reset_index()
fig = px.choropleth(region_data,
                    locations='geoName',
                    locationmode = 'country names',
                    color= keyword,
                    title = f'<b>Search Interest for {keyword} by country</b>',
                    color_continuous_scale=  'Blues',
                    width=1500,
                    height=600 )
fig.show()
```

...



Code: Creates a world map showing country-wise search interest for the selected keyword.

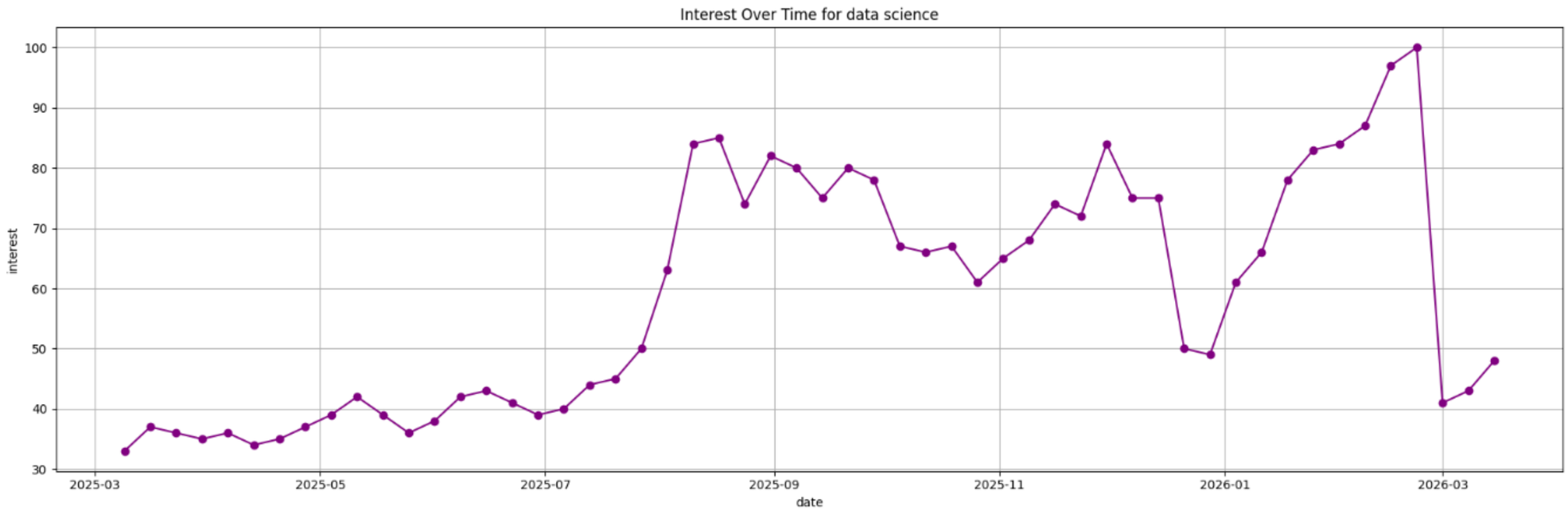
Graph: The world map shows the global distribution of search interest for the keyword. Darker shades represent higher search interest, while lighter shades indicate lower interest, helping to easily identify the countries where the keyword is most popular.

WORK ON LIBRARYS

Time Wise Interest

```python
time_df = pytrends.interest_over_time()
```

```python
plt.figure(figsize = (18,6))
plt.plot(time_df.index , time_df[keyword] , marker ='o' , color = 'purple')
plt.title(f'Interest Over Time for {keyword}')
plt.xlabel("date")
plt.ylabel("interest")
plt.grid(True)
plt.tight_layout()
plt.show()
```



Code: Extracts the keyword's search interest over time and plots it against the date to analyze trends.

Graph: Shows the trend of keyword searches over time, highlighting periods of increase and decrease in interest. 📈

WORK ON LIBRARYS

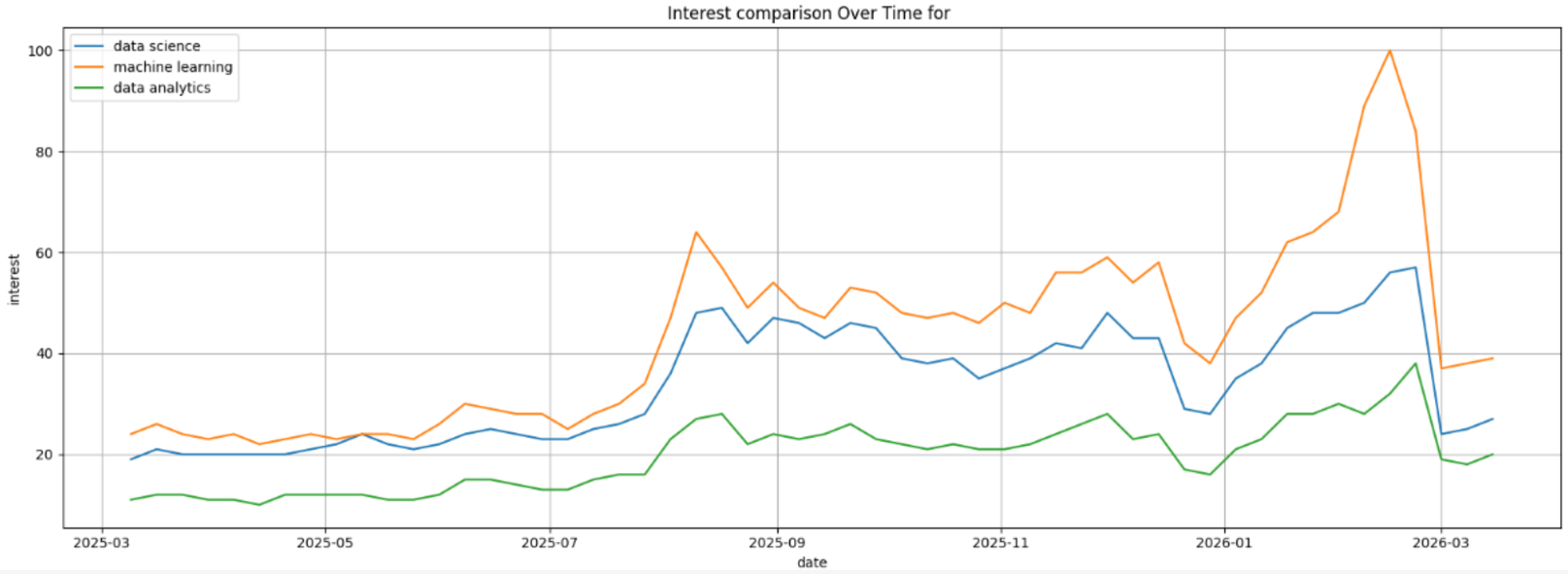
Multiple keywords compare

```
kw_list = ['data science', 'machine learning' , 'data analytics' ]
pytrends.build_payload(kw_list , cat=0,timeframe='today 12-m',geo='',gprop='')
```

```
compare_df = pytrends.interest_over_time()


plt.figure(figsize = (16,6))
for kw in kw_list:
  plt.plot(compare_df.index , compare_df[kw] , label = kw)

plt.title(f'Interest comparison Over Time for ')
plt.xlabel("date")
plt.ylabel("interest")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```



Code: The code collects search trend data for the keywords data science, machine learning, and data analytics over the last 12 months and plots a line graph to compare their interest over time.

Graph: It shows that machine learning has the highest search popularity, followed by data science, while data analytics has comparatively lower interest. 📈

EXECUTIVE SUMMARY

This project studies the popularity of data-related keywords using Google search trend data. It focuses on **Data Science, Machine Learning, and Data Analytics** to understand how people's interest in these topics has changed over time and which one is searched the most. The project collects data from the last 12 months and presents it using graphs so the comparison is easy to understand. The analysis compares these keywords and shows their trends over time. The results indicate that **Machine Learning has the highest search interest**, followed by **Data Science,** while **Data Analytics** has **lower but slowly increasing interest**. This helps show current technology trends and the growing importance of data-related skills. 📊

CONCLUSION

This project analyzed Google search trends for **Data Science, Machine Learning, and Data Analytics** over the last 12 months. The results show that **Machine Learning has the highest search interest,** followed by Data Science, while **Data Analytics** has **lower but gradually increasing popularity**. Overall, the analysis helps understand current technology trends and highlights the growing demand for data-related skills. 📊

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